

23L-6006 Mudassar hussain
6B2

Sample

```
#include <stdio.h>
#include<string.h>
struct fileTable
{
    char name[20];
    int sb, nob;
} ft[30];

void main()
{
    int i, j, n;
    char s[20];

    printf("Enter no of files :");
    scanf("%d", &n);

    for (i = 0; i < n; i++)
    {
        printf("\nEnter file name %d :", i + 1);
        scanf("%s", ft[i].name);

        printf("Enter starting block of file %d :", i + 1);
        scanf("%d", &ft[i].sb);

        printf("Enter no of blocks in file %d :", i + 1);
        scanf("%d", &ft[i].nob);
    }

    printf("\nEnter the file name to be searched -- ");
    scanf("%s", s);

    for (i = 0; i < n; i++)
        if (strcmp(s, ft[i].name) == 0)
            break;

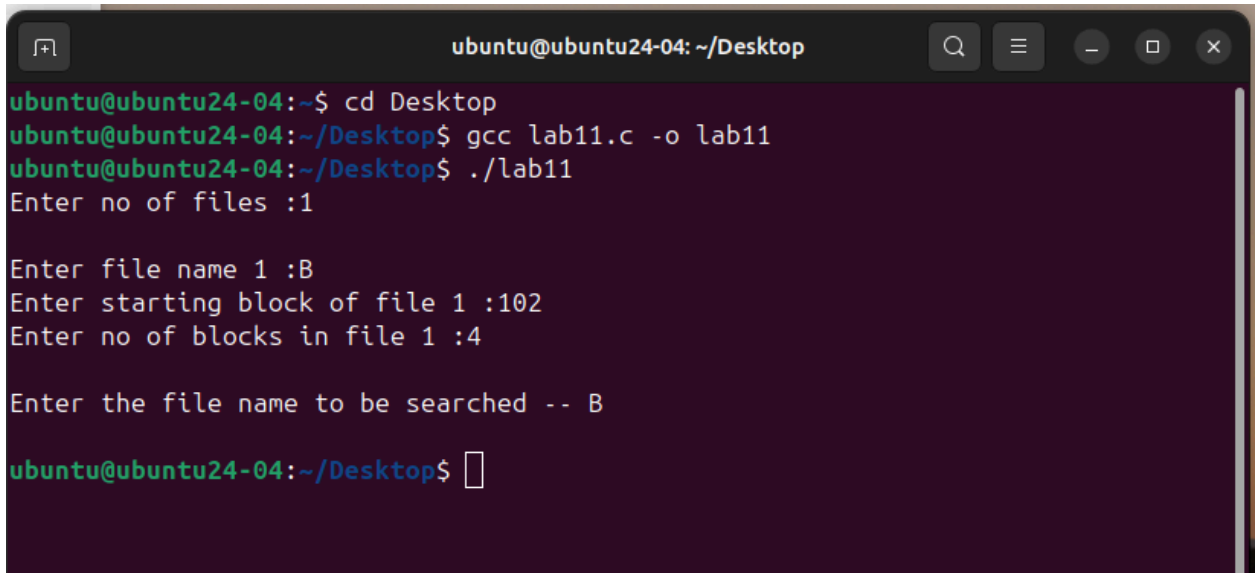
    if (i == n)
        printf("\nFILE NAME OCCUPIED\n");
```

```

printf("\n%s\t\t%d\t\t%d\t", ft[i].name, ft[i].sb, ft[i].nob);

for (j = 0; j < ft[i].nob; j++)
    printf("%d, ", ft[i].sb + j);
}

```



```

ubuntu@ubuntu24-04: ~/Desktop
ubuntu@ubuntu24-04:~$ cd Desktop
ubuntu@ubuntu24-04:~/Desktop$ gcc lab11.c -o lab11
ubuntu@ubuntu24-04:~/Desktop$ ./lab11
Enter no of files :1

Enter file name 1 :B
Enter starting block of file 1 :102
Enter no of blocks in file 1 :4

Enter the file name to be searched -- B

ubuntu@ubuntu24-04:~/Desktop$

```

Q1

```

#include <iostream>
#include <cstring>
using namespace std;

// Structure for each disk block
struct block
{
    int bno;
    block *next;
};

// Structure for file table
struct fileTable
{
    char name[20];
    int nob;    // number of blocks

```

```

    block *sb;    // starting block
} ft[30];

int main()
{
    int n; // number of files

    cout << "Enter number of files: ";
    cin >> n;

    for (int i = 0; i < n; i++)
    {
        cout << "\nEnter file name: ";
        cin >> ft[i].name;

        cout << "Enter number of blocks: ";
        cin >> ft[i].nob;

        block *temp, *newBlock;

        for (int j = 0; j < ft[i].nob; j++)
        {
            newBlock = new block;

            cout << "Enter block number " << j + 1 << ": ";
            cin >> newBlock->bno;
            newBlock->next = NULL;

            if (j == 0)
            {
                ft[i].sb = newBlock; // first block
                temp = newBlock;
            }
            else
            {
                temp->next = newBlock; // link previous block
                temp = newBlock;
            }
        }
    }

    // Display file allocation table
    cout << "\n\nFILE NAME\tNO OF BLOCKS\tBLOCKS OCCUPIED\n";
    cout << "-----\n";

```

```

for (int i = 0; i < n; i++)
{
    cout << ft[i].name << "\t\t" << ft[i].nob << "\t\t";

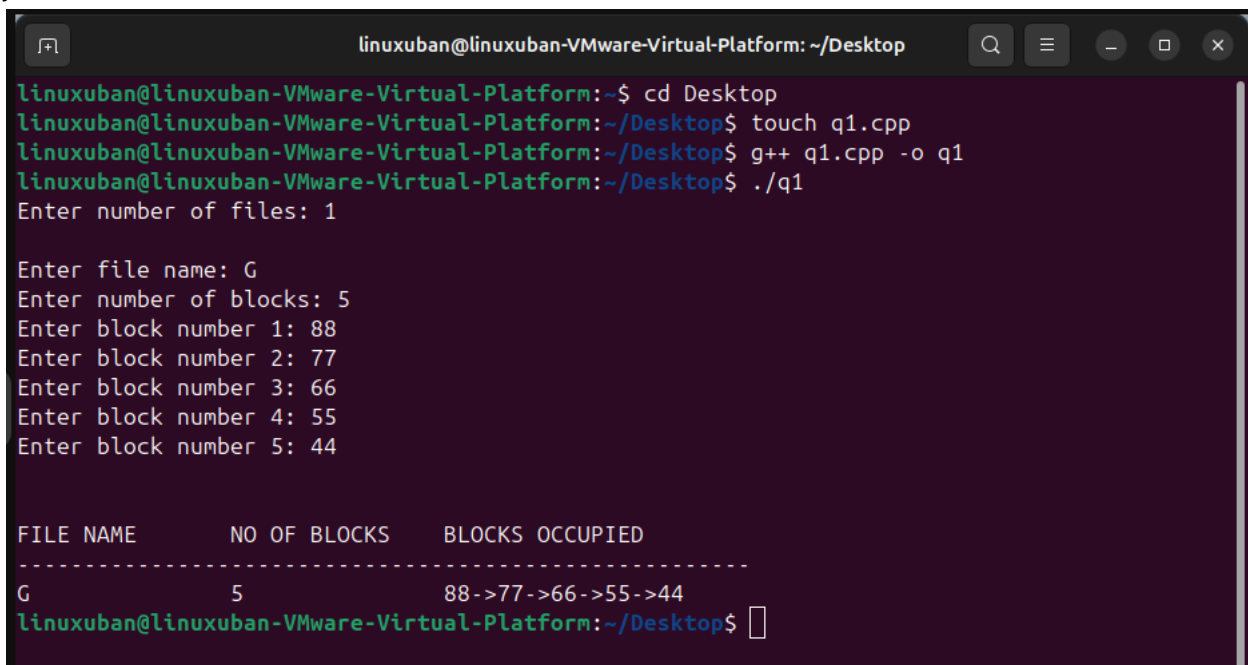
    block *temp = ft[i].sb;

    while (temp != NULL)
    {
        cout << temp->bno;
        if (temp->next != NULL)
            cout << "->";
        temp = temp->next;
    }

    cout << endl;
}

return 0;
}

```



```

linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop
linuxuban@linuxuban-VMware-Virtual-Platform:~$ cd Desktop
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop$ touch q1.cpp
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop$ g++ q1.cpp -o q1
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop$ ./q1
Enter number of files: 1

Enter file name: G
Enter number of blocks: 5
Enter block number 1: 88
Enter block number 2: 77
Enter block number 3: 66
Enter block number 4: 55
Enter block number 5: 44

FILE NAME      NO OF BLOCKS   BLOCKS OCCUPIED
-----
G              5              88->77->66->55->44
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop$

```

Q2

```

#include <iostream>
#include <cstring>
using namespace std;

// Structure for file table
struct fileTable
{
    char name[20];
    int nob;          // Number of blocks
    int blocks[30];   // Indexed blocks
} ft[30];

int main()
{
    int n; // Number of files

    cout << "Enter number of files: ";
    cin >> n;

    // Input file details
    for (int i = 0; i < n; i++)
    {
        cout << "\nEnter file name: ";
        cin >> ft[i].name;

        cout << "Enter number of blocks: ";
        cin >> ft[i].nob;

        cout << "Enter block numbers:\n";
        for (int j = 0; j < ft[i].nob; j++)
        {
            cout << "Block " << j + 1 << ": ";
            cin >> ft[i].blocks[j];
        }
    }

    // Display Indexed File Allocation Table
    cout << "\n\nFILE NAME\tNO OF BLOCKS\tBLOCKS OCCUPIED\n";
    cout << "-----\n";

    for (int i = 0; i < n; i++)
    {
        cout << ft[i].name << "\t\t" << ft[i].nob << "\t\t";
    }
}

```

```

        for (int j = 0; j < ft[i].nob; j++)
        {
            cout << ft[i].blocks[j];
            if (j != ft[i].nob - 1)
                cout << ", ";
        }

        cout << endl;
    }

    return 0;
}

```

The terminal window shows the compilation and execution of a C++ program. The user enters the number of files (2), then for each file, the file name, number of blocks, and the specific block numbers are entered. The program then displays a table of the entered data.

```

linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop$ g++ q1.cpp -o q1
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop$ ./q1
Enter number of files: 2

Enter file name: A
Enter number of blocks: 3
Enter block numbers:
Block 1: 10
Block 2: 20
Block 3: 30

Enter file name: B
Enter number of blocks: 4
Enter block numbers:
Block 1: 40
Block 2: 50
Block 3: 60
Block 4: 70

FILE NAME      NO OF BLOCKS   BLOCKS OCCUPIED
-----
A              3              10, 20, 30
B              4              40, 50, 60, 70
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop$ 

```